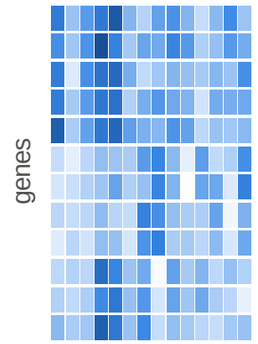
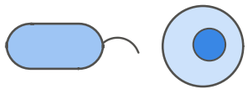
# Adding a softmax and a multi-step walk to CLR's normalization recovers large *E. coli* regulons better (16% over CLR), but loses resolution on tight protein complexes

## CLR (Faith et al. 2007)

### 1 Expression compendium

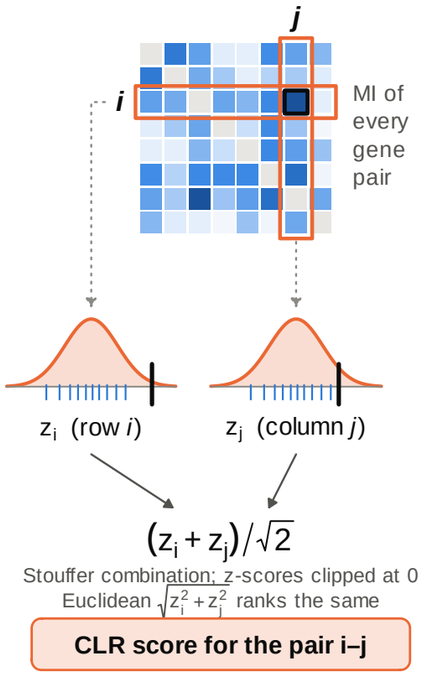



M3D, EcoMAC microarrays — K562 Perturb-seq

### 2 Mutual information, z-scored against each gene's own background

existing method



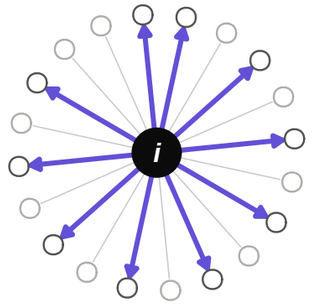

Stouffer combination; z-scores clipped at 0
Euclidean $\sqrt{z_i^2 + z_j^2}$ ranks the same

**CLR score for the pair i–j**

*Published CLR stops here: a score for each gene pair*

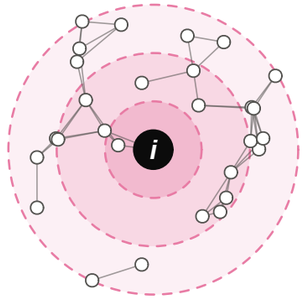
## This work: steps taken from attention

### 3 Softmax with per-gene temperature




~10 effective partners per gene, set in advance

### 4 Multi-step walk, number of steps chosen without labels




steps t (log scale)




t = 9: chosen on held-out labs

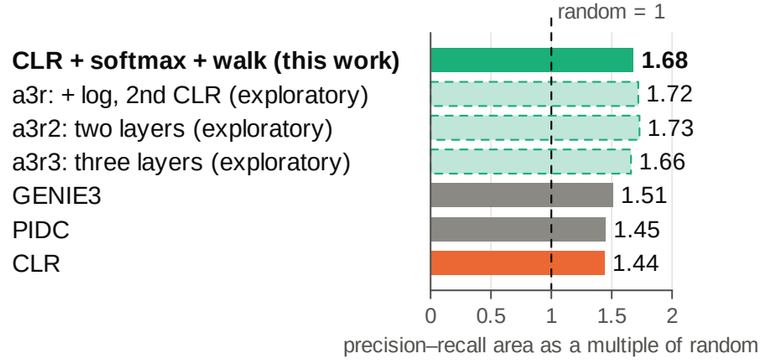
**No learned parameters**

**Transformer analogy:** MI ≈ query·key score; CLR z-score ≈ scaled logit; softmax ≈ softmax; walk ≈ repeated attention with residual

## 5 What it recovers

### E. coli regulons (experimentally supported, RegulonDB)

regulon = genes controlled by one transcription factor (TF)




*per regulon: tie with CLR and GENIE3;*
*ahead in 1,000/1,000 resamplings vs CLR*

### Protein complexes (K562 CORUM; mostly ribosome): this work loses 1.8×

complex = proteins that bind as one machine

